The M/G/1 queue

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February 16, 2021

Abstract

This note presents an overview of the M/G/1 queue.

1 Introduction

Our discussion so far has assumed that both arrivals and departures are Markov. While Markov arrivals are quite reasonable in practice, especially given that the merger of two Markov arrival streams results also in a Markov stream, departures are usually far from Markov. For example, in the Internet, packets are usually bimodally distributed in length, with large data packets and short acknowledgments. Since these are transmitted over constant-rate links, the departure process is also bimodal, rather than Markov. Specifically, the time between two departures in not drawn from an exponential distribution; instead, it is drawn from a bimodal distribution. Thus, if we 'look' at the queue at a particular point in time, the expected remaining service time for the packet currently in service is not memoryless, which is only true for the exponential distribution. This makes the analysis more complex.

In this note, we will not derive the formulae for the M/G/1 queue, and instead only present the results.

2 Parameters

The three parameters that are used to describe an M/G/1 queue are:

- λ, the mean arrival rate
- μ, the mean service rate. Note that if the service time is denoted by S, then $\mathbb{E}(S) = 1/\mu$
- C_S^2, which is the square of the coefficient of variation of the service time. $C_S^2 = \frac{\mathbb{E}(S^2)}{(\mathbb{E}(S))^2} - 1$

3 Expected time in system

Given these parameters, then the expected value of T, the time in the system for a typical job, i.e., the sum of the time waiting for service and the service time, is given by:

$$\mathbb{E}(T) = \frac{1}{\mu}(1 + \frac{\rho(1 + C_s^2)}{2(1 - \rho)}) \tag{1}$$

4 Expected number of jobs in the system

The expected number of jobs in the system, denoted $\mathbb{E}(N)$ is given by:

$$\mathbb{E}(N) = \lambda\,\mathbb{E}(T) = \rho + \frac{\rho^2(1 + C_s^2)}{2(1 - \rho)} \tag{2}$$

This is also known as the **Pollaczek-Khintchine formula**. Note that the number of jobs in the queue depends not only on λ and μ, as in the M/M/1 queue, but also on C_S^2.

- If this value is much larger than 1, then the number of jobs in the system is much greater than in an M/M/1 queue with the same arrival and departure rates.

- If $C_S^2 = 1$, which is the case for Markov departurs, then $\mathbb{E}(N)$ reduces to $\frac{\rho}{1-\rho}$, which is the formula for the number of jobs in the system for the M/M/1 queue.
- If C_S^2 is smaller than 1, then the M/M/1 queue can be used to approximate the system, though the results will be somewhat conservative.

Example 1: Suppose that the service time distribution of a queue is described by $C_S^2 = 5$. Compare the time in the system for this queue with that of a M/M/1 queue with the same mean arrival and departure rates.

Answer: From Eq. 1, for the M/M/1 queue, where $C_S^2 = 1$,

$$\mathbb{E}(T) = \frac{1}{\mu}(1 + \frac{\rho(1 + C_s^2)}{2(1-\rho)})$$

$$= \frac{1}{\mu}(1 + \frac{\rho}{(1-\rho)})$$

$$= \frac{1}{\mu}(\frac{1}{(1-\rho)})$$

When $C_S^2 = 5$, then

$$\mathbb{E}(T) = \frac{1}{\mu}(1 + \frac{\rho(6)}{2(1-\rho)})$$

$$\mathbb{E}(T) = \frac{1}{\mu}(\frac{1+2\rho}{(1-\rho)})$$

Taking their ratio, we see that for this queue, the mean time in the system exceeds that of a corresponding M/M/1 queue by a factor of $1+2\rho$.